PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 1 • Chapter OL-T25 Classified

Cross-session • chapter-organised candidate

9 classified items • 23 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 1 • Unit 1 • Chapter OL-T25 • 9 items • 23 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Solving Quadratic Equations

Unit 1 • 23 marks • 9 item(s)

June 2024 | 4MA1-2H Q11 | 3 marks

Monic quadratic roots by factors

June 2023 | 4MA1-1H Q6 | 3 marks

Monic quadratic roots by factors

November 2024 | 4MA1-1H Q8 | 3 marks

Monic quadratic roots by factors

June 2022 | 4MA1-2H Q11 | 3 marks

Monic quadratic roots by factors

January 2022 | 4MA1-1H Q6 | 3 marks

Monic quadratic roots by factors

January 2020 | 4MA1-2H Q7 | 3 marks

Monic quadratic roots by factors

January 2021 | 4MA1-1H Q9 | 1 marks

Monic quadratic roots by factors

June 2021 | 4MA1-1H Q9 | 3 marks

Quadratic rearranged before factorising

June 2021 | 4MA1-2H Q9 | 1 marks

Monic quadratic roots by factors

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Question 22

4MA1-2H Q11

ORIGINAL QUESTION
3 marks

11 (i) Factorise $x^2 + 9x - 22$

....
(2)

(ii) Hence, solve $x^2 + 9x - 22 = 0$

....
(1)

Worked Solution 22

Factorise, hence solve

MODULAR UNIT 1

3 marks

Factorise

$$x^2 + 9x - 22 = (x - 2)(x + 11).$$

Roots

$$(x - 2)(x + 11) = 0.$$

Final answer

$$(x - 2)(x + 11); \quad x = 2, -11$$

Question 05

4MA1-1H Q6

ORIGINAL QUESTION
3 marks

(d) (i) Factorise $x^2 + 9x - 22$

....
(2)

(ii) Hence solve $x^2 + 9x - 22 = 0$

....
(1)

Worked Solution 05

Factorise then solve

MODULAR UNIT 1

3 marks

Factorise

$$x^2 + 9x - 22 = (x + 11)(x - 2).$$

Use the zero-product rule

$$x + 11 = 0 \text{ or } x - 2 = 0.$$

Final answer

$$x = -11 \text{ or } x = 2$$

Original question 07

4MA1-1H Q8 M01, M02

MODULAR UNIT 1
3 marks

8 (a) (i) Factorise $x^2 + 5x - 24$

....
(2)

(ii) Hence, solve $x^2 + 5x - 24 = 0$

....
(1)

Worked solution 07

Factorise a quadratic / Solve the factorised equation

MODULAR UNIT 1
3 marks

M01 – Factorise a quadratic

Find two numbers

The product is -24 and the sum is 5, so use 8 and -3.

Write the factors

$$x^2 + 5x - 24 = (x + 8)(x - 3).$$

Final answer

(x+8)(x-3)

M02 – Solve the factorised equation

Set each factor to zero

$$x + 8 = 0 \text{ or } x - 3 = 0.$$

State both roots

$$x = -8 \text{ or } x = 3.$$

Final answer

$$x = -8 \text{ or } x = 3$$

Original question 21

4MA1-2H Q11 Q11(i), Q11(ii)

ORIGINAL QUESTION
3 marks

11 (i) Factorise $x^2 + 5x - 24$

....

(2)

(ii) Hence, solve $x^2 + 5x - 24 = 0$

....

(1)

Worked solution 21

Chapter: Factorising; Solving Quadratic Equations

MODULAR UNIT 1
3 marks

Q11(i) – Chapter: Factorising

Family: Factorise a monic quadratic • Variant: Negative constant monic quadratic

Method
The required numbers have product -24 and sum 5, so they are 8 and -3.
Why: Match the constant term and middle coefficient.

Method
Therefore $x^2 + 5x - 24 = (x - 3)(x + 8)$.
Why: Write the monic quadratic as two linear factors.

Final answer
factorisation: $(x - 3)(x + 8)$

Q11(ii) – Chapter: Solving Quadratic Equations

Family: Solve a quadratic by factorisation • Variant: Monic quadratic roots by factors

Method
From $(x - 3)(x + 8) = 0$, $x = 3$ or $x = -8$.
Why: Use the zero-product rule on the factors from part (i).

Final answer
roots: $x = 3$ or $x = -8$

Original question 05

4MA1-1H Q6 Q6(b)(i), Q6(b)(ii)

ORIGINAL QUESTION
3 marks

(b) (i) Factorise $y^2 - 2y - 35$

....
(2)

(ii) Hence, solve $y^2 - 2y - 35 = 0$

....
(1)

Worked solution 05

Chapter: Factorising; Solving Quadratic Equations

MODULAR UNIT 1
3 marks

Q6(b)(i) – Chapter: Factorising

Family: Factorise a non-monic or special quadratic • Variant: Factorise as the first part of a factor-then-solve chain

Method
We need two numbers with product -35 and sum -2: -7 and $+5$.
Why: The constant term is -35 and the x-coefficient is -2.

Method
$y^2-2y-35 = (y-7)(y+5)$.
Why: Expand the brackets to check.

Final answer
factorisation: $(y-7)(y+5)$

Q6(b)(ii) – Chapter: Solving Quadratic Equations

Family: Solve a quadratic by factorisation • Variant: Monic quadratic roots by factors

Method
$(y-7)(y+5)=0$, so $y=7$ or $y=-5$.
Why: Use the zero-product rule; this follows the factorisation in part (i).

Final answer
roots: $y = 7$ or $y = -5$

(b) Solve $x^2 - 3x - 40 = 0$

Show clear algebraic working.

....

(3)

Worked solution

January 2020 | Question 7 | 3 marks

Family: Solve a quadratic by factorisation • Variant: Monic quadratic roots by factors

Method / working

1. $x^2-3x-40=(x-8)(x+5)$.
2. Set each factor to zero.
3. Roots are $x=8$ and $x=-5$.

Final answer: $x=8$ or $x=-5$

January 2021 | 4MA1-1H | Question 9 | 1 marks

Chapter OL-T25: Solving Quadratic Equations

(b) (i) Factorise $x^2 + 5x - 36$

....

(2)

(ii) Hence, solve $x^2 + 5x - 36 = 0$

....

(1)

Worked solution

January 2021 | Question 9 | 1 marks

Family: Solve a quadratic by factorisation • Variant: Monic quadratic roots by factors

Method / working

1. Set each factor to zero: $x+9=0$ or $x-4=0$.

Final answer: $x=-9$ or $x=4$

9 (a) Factorise fully $25a^4c^7d + 45a^9c^3h$

....
(2)

(b) Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

$x =$ ....
(3)

Worked solution

June 2021 | Question 9 | 3 marks

Family: Solve a quadratic by factorisation • Variant: Quadratic rearranged before factorising

Method / working

1. Expand both sides: $4x^2+20x+25 = 4x^2+4x-3$.
2. Collect x terms and constants: $16x = -28$.
3. Hence $x = -28/16 = -7/4 = -1.75$.

Final answer: $x = -7/4$

June 2021 | 4MA1-2H | Question 9 | 1 marks

Chapter OL-T25: Solving Quadratic Equations

9 (i) Factorise $x^2 + 2x - 24$

....
(2)

(ii) Hence solve $x^2 + 2x - 24 = 0$

....
(1)

Worked solution

June 2021 | Question 9 | 1 marks

Family: Solve a quadratic by factorisation • Variant: Monic quadratic roots by factors

Method / working

1. Set each factor to zero: $x+6=0$ or $x-4=0$.

Final answer: $x=-6$ or 4